Week 12 Self-Assessment

Research Design Framework

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1 Instructions

This is the most important quiz of the term. Week 12 introduces ALL the terminology you need for your Research Plan. Make sure you can answer these questions confidently.

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2 Part 1: Variables

2.1 Question 1: IV and DV Definitions

The variable that the researcher **manipulates or selects** is called the:

- (A) Dependent variable
- (B) Independent variable
- (C) Control variable
- (D) Confounding variable

The variable that the researcher **measures** is called the:

- (A) Dependent variable
- (B) Independent variable
- (C) Control variable
- (D) Confounding variable

2.2 Question 2: Identifying Variables

A researcher compares memory scores between participants who slept 8 hours versus 4 hours.

What is the IV?

- (A) Sleep duration (8 hrs vs 4 hrs)
- (B) Memory score
- (C) Participant age
- (D) Time of testing

What is the DV?

- (A) Sleep duration
- (B) Memory score

- (C) Number of participants
- (D) Room temperature

2.3 Question 3: DataLab Variables

In our **Room A vs Room B** comparison of reaction times:

The IV is:

- (A) Room (A vs B)
- (B) Reaction time
- (C) Participant
- (D) Time of day

The DV is:

- (A) Room assignment
- (B) Reaction time
- (C) Number of trials
- (D) Catch distance

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3 Part 2: Design Types

3.1 Question 4: Design Definitions

Match each design to its definition:

Different participants in each condition =

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

Same participants tested in all conditions =

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

Participants **paired on characteristics**, one from each pair in each condition =

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

No manipulation, just measuring relationships =

- (A) Between-subjects

- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

3.2 Question 5: DataLab Design Examples

What type of design is **Room A vs Room B** (different students in each room)?

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

What type of design is **Pre-mood vs Post-mood** (same students measured twice)?

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

What type of design is **Confidence rating vs Throwing accuracy** (measuring relationship)?

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

3.3 Question 6: No IV

In a correlational design, there is no IV. TRUE / FALSE

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4 Part 3: Hypotheses

4.1 Question 7: Hypothesis Types

A hypothesis that predicts a difference but NOT the direction (e.g., "scores will differ") is:

- (A) One-tailed
- (B) Two-tailed
- (C) Null
- (D) Alternative

A hypothesis that predicts the direction (e.g., "Group A will score higher than Group B") is:

- (A) One-tailed
- (B) Two-tailed
- (C) Null
- (D) Alternative

4.2 Question 8: Null vs Experimental

The **null hypothesis** predicts:

- (A) A significant effect
- (B) No effect or no difference
- (C) The direction of the effect
- (D) That the study will fail

The **experimental hypothesis** (also called the alternative hypothesis) predicts:

- (A) An effect or difference exists
- (B) No effect or difference
- (C) The study is flawed
- (D) Random variation

4.3 Question 9: Writing Hypotheses

Which is a correctly written **one-tailed** hypothesis?

```
opts_hyp <- c(
  "Reaction times will differ between Room A and Room B",
  answer = "Participants in Room A will have faster reaction times than
participants in Room B",
  "There will be no difference in reaction times between rooms",
  "Reaction times may or may not differ between rooms"
)
```

- (A) Reaction times will differ between Room A and Room B
- (B) Participants in Room A will have faster reaction times than participants in Room B
- (C) There will be no difference in reaction times between rooms
- (D) Reaction times may or may not differ between rooms

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5 Part 4: Test Selection

5.1 Question 10: Design to Test Matching

Fill in the correct test for each design:

Between-subjects design → ____________________

Within-subjects design → ________________

Correlational design → ________

5.2 Question 11: Test Selection MCQ

A researcher compares anxiety levels before and after a relaxation intervention (same participants).

Which test should they use?

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Mann-Whitney U

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6 Part 5: Descriptive Statistics

6.1 Question 12: Calculation

Calculate the **mean** of these scores: 10, 15, 20, 25, 30

Mean = __

6.2 Question 13: Standard Deviation

The standard deviation tells us:

- (A) The middle value
- (B) The most common value
- (C) How spread out values are around the mean
- (D) The difference between highest and lowest

6.3 Question 14: APA Format

In APA format, how do you report a mean of 25.50 and standard deviation of 4.30?

- (A) Mean = 25.50, SD = 4.30
- (B) mean: 25.50, sd: 4.30
- (C) M = 25.50, SD = 4.30
- (D) Average = 25.50 (4.30)

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7 Part 6: MCQ Exam Practice

7.1 Investigation 1

A researcher wants to know if background music affects concentration. Half the participants complete a task with music playing, half complete it in silence. Concentration is measured by number of errors.

1. **Design**:

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV**:

- (A) Music condition (music vs silence)
- (B) Number of errors

- (C) Concentration level
- (D) No IV

3. **DV**:

- (A) Music condition
- (B) Number of errors
- (C) Background noise level
- (D) Participant preference

4. **Appropriate test**:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson’s r
- (D) Chi-square

7.2 Investigation 2

A researcher measures students’ anxiety before an exam and their exam performance to see if they are related.

1. **Design**:

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV**:

- (A) Anxiety level
- (B) Exam performance
- (C) Student year
- (D) No IV

3. **DV**:

- (A) Anxiety level
- (B) Exam performance
- (C) Both variables are measured (no DV)
- (D) Student motivation

4. **Appropriate test**:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson’s r
- (D) One-way ANOVA

7.3 Investigation 3

Participants rate their mood before and after watching a comedy video.

1. **Design**:

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV**:

- (A) Time (before vs after video)
- (B) Mood rating
- (C) Video content
- (D) No IV

3. **DV**:

- (A) Time point
- (B) Mood rating
- (C) Video length
- (D) Participant age

4. **Appropriate test**:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Mann-Whitney U

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8 Summary Checklist

Before moving on, make sure you can:

- ☐ Define IV (independent variable) and DV (dependent variable)
- ☐ Identify IV and DV in any research scenario
- ☐ Define all four design types (between, within, matched, correlational)
- ☐ Recognise when there is no IV (correlational design)
- ☐ Write one-tailed and two-tailed hypotheses
- ☐ Distinguish experimental from null hypotheses
- ☐ Select the correct test for each design type
- ☐ Calculate mean and interpret SD
- ☐ Report statistics in APA format

! Research Plan Connection

You need ALL of these skills for your Research Plan (due Week 16). If you're unsure about any concept, review the Week 12 lecture materials before continuing.